A QUOTIENT OF A SMOOTH SPACE BY A FLAT FINITELY PRESENTED GROUP IS SMOOTH OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The theorem that a flat groupoid in algebraic spaces gives an algebraic stack. Invoked at the start of the counted proof (source lines 7709-7711) as a cross-chapter pointer. Makes the quotient stack algebraic.
- The statement that the covering of the quotient is flat and locally of finite presentation. Invoked inside the counted proof (source lines 7712-7718) as a cross-chapter pointer. Supplies the base change used to apply descent for smoothness.
- The descent statement for smoothness along a flat finitely presented surjection. Invoked at the end of the counted proof (source lines 7719-7724) as a cross-chapter pointer. Concludes smoothness of the quotient over the base.

Source reference. Original reference criteria-lemma-flat-quotient-flat-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3877–3877. Navigation pointer only: the counted text refers to the Stacks item criteria-lemma-flat-quotient-flat-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference criteria-theorem-flat-groupoid-gives-algebraic-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3850–3850. Navigation pointer only: the counted text refers to the Stacks item criteria-theorem-flat-groupoid-gives-algebraic-stack, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's

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Source reference. Original reference spaces-descent-lemma-smooth-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7724–7724. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-smooth-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $X \to Y$ be a smooth morphism of algebraic spaces. Let G be a group algebraic space over Y which is flat and locally of finite presentation over Y. Let G act on X over Y. Then the quotient stack $[X/G]$ is smooth over Y.*

Proof. The quotient $[X/G]$ is an algebraic stack by Criteria for Representability, Theorem criteria-theorem-flat-groupoid-gives-algebraic-stack. The smoothness of $[X/G]$ over Y follows from the fact that smoothness descends under fppf coverings: Choose a surjective smooth morphism $U \to [X/G]$ where U is a scheme. Smoothness of $[X/G]$ over Y is equivalent to smoothness of U over Y. Observe that $U \times_{[X/G]} X$ is smooth over X and hence smooth over Y (because compositions of smooth morphisms are smooth). On the other hand, $U \times_{[X/G]} X \to U$ is locally of finite presentation, flat, and surjective (because it is the base change of $X \to [X/G]$ which has those properties for example by Criteria for Representability, Lemma criteria-lemma-flat-quotient-flat-presentation). Therefore we may apply Descent on Spaces, Lemma spaces-descent-lemma-smooth-permanence. □